Received: 9 May 2025 | Revised: 13 October 2025 | Accepted: 30 October 2025

DOI: 10.1111/test.70022

TECHNOLOGY SPOTLIGHT

WILEY

How to be "Choosy": Wrangling big datasets for the classroom

Michelle H. Wilkerson[1] | **Tim Erickson**[2] | **Hollylynne S. Lee**[3] | **William Finzer**[4]

[1]Berkeley School of Education, University of California, Berkeley, California, USA

[2]eeps media, Oakland, California, USA

[3]College of Education, North Carolina State University, Raleigh, North Carolina, USA

[4]Concord Consortium, Concord, Massachusetts, USA

Correspondence
Michelle H. Wilkerson, Berkeley School of Education, University of California, 2121 Berkeley Way West, Berkeley, CA, USA.
Email: mwilkers@berkeley.edu

Funding information
National Science Foundation, Grant/Award Number: IIS-1900606

Abstract

Educators are being encouraged to teach with "big" datasets that have more cases and attributes than are typically used in the classroom. When introduced carefully, these types of datasets can allow students to engage in complex and self-directed reasoning, develop data management and inquiry skills, and experience data analysis in a way that is more authentic to professional practice. However, "big data" is also often unwieldy. It can overwhelm students, overload their software tools, and interfere with planful analysis. How can we make such datasets manageable? This paper presents pedagogical strategies and technical methods for educators, educational designers, and young investigators to reduce the number of cases or attributes in a large dataset in ways that do not unduly compromise their analyses. We illustrate these strategies with interactive demonstrations in two freely-available open source tools: the Choosy plugin in CODAP, and a Python Jupyter notebook.

KEYWORDS

data science education, data wrangling, educational datasets, teaching statistics

1 | INTRODUCTION

Data is at the core of meaningful learning experiences in statistics and data science. Increasingly, educators are encouraged to expose students to working with "big" or "messy" datasets (Gould, 2010; Rosenberg et al., 2020; Schultheis & Kjelvik, 2020), many of which are open and publicly available. Scholars have presented a number of compelling arguments for why exposing learners to large, public datasets is important, and the movement toward more complex data sources is also reflected in emerging curricular guidelines and standards documents (Krishnamurthi et al., n.d.).

There is less guidance, however, around how these types of datasets can be wrangled into an appropriate size and complexity for a particular classroom or task, without seriously compromising the benefits and richness of the data. The goal of this paper is to illustrate pedagogically informed strategies for "wrangling" large datasets to a reasonable size by selectively reducing the number of cases and/or reducing the number of attributes. These include three strategies for reducing the number of cases: *random selection*, *purposeful selection by attribute*, and *build-your-own attributes*. We also present three strategies for reducing the number of attributes: *thematic selection*, *mathematics-driven selection*, and *question-driven selection*. We present each strategy, with attention to their pedagogical alignments, to help educators and curriculum designers evaluate when and how each should be applied in order to support specific intended learning goals and experiences. While these were developed and will be discussed in the context of precollegiate statistics and data science education, we suspect they can be useful for anyone seeking to reduce the size of a dataset while maintaining their pedagogical benefits.

In the following sections, we demonstrate how these six strategies can be executed using key features of two free, open-source platforms commonly used in data and statistics education, the Common Online Data Analysis Platform (CODAP) and Python Jupyter notebooks. These represent two broad classes of tools that are especially popular in educational settings—visual and table-based tools (e.g., Tuva, DataClassroom, and Tableau), and script-based tools (R and RStudio, pyret), respectively. Example interactive CODAP documents and Python Jupyter notebooks can be accessed online (for double-anonymized peer review, we have provided links to anonymized versions of these materials below), when each dataset is introduced, so readers can get first-hand experience exploring the technical and pedagogical affordances of each tool in preparing datasets for and with students. At the end of the article and in Supporting Information S1, "Quick Guide," we summarize each data reduction strategy, its pedagogical benefits and drawbacks, and its connections to major standards in statistics and data science (e.g., Guidelines for Assessment and Instruction in Statistics Education "GAISE"; Bargagliotti et al., 2020; and the International Data Science in Schools Project "IDSSP"; IDSSP Curriculum Team, 2019).

1.1 | What makes a dataset (too) big?

There are a variety of ideas about what makes data "big"—for example, data's increased role in everyday life (Lane et al., 2013, as cited in Ainley et al., 2015), or the complexity, volume, heterogeneity, sourcing, or processing of a given dataset (Favaretto et al., 2020). For the purposes of this paper, we take a more practical view: A dataset may include a large number of *cases*, or specific observations of a phenomenon of interest. When we display this type of large dataset as a table, we might describe the table as "long," or as having many "rows," "records," or "observations." Similarly, a dataset may include a large number of *attributes*, which measure or describe different characteristics for each case. When we display these datasets as a table they appear "wide," with many "columns," "variables," or "dimensions." While different readers may be familiar with these or other terms when describing datasets, for consistency we will use the terms *cases* and *attributes* throughout this paper.

These two ideas of "big" as having many *cases* or many *attributes* each allow for different types of investigation:

1. Having more cases in a dataset improves the statistical power and reliability of an analysis, and allows for detailed exploration of distributions, comparisons, and uncertainty through tasks such as "growing samples" (Ben-Zvi et al., 2015). Consider this: if you randomly select 1000 people from the entire United States, on average, only two of them will be from the state of Vermont. How could you make any meaningful conclusion about how Vermont compares with other states?
2. Having more attributes in a dataset allows more exploration of multivariate relationships, allowing students to explore different, but related questions and to leverage contextual knowledge (Kazak et al., 2023; Wilkerson & Laina, 2018).

Of course, as is often the case when exploring public datasets, a dataset may be "big" in both ways at the same time.

One problem is that many datasets that could be especially interesting or powerful for students have become *too big*. It is now easy to find data with dozens, even hundreds of attributes or with tens of thousands of cases. Such datasets can allow for a variety of rich, student-directed investigations. But too many cases and attributes can also present conceptual and logistical challenges that are difficult to overcome in the classroom. Conceptually, students need scaffolding to approach data with guiding questions, hypotheses, and expectations for how they will pursue analysis and what they might find (Arnold et al., 2022). This becomes difficult when there are so many potential avenues to explore. Logistically, students and teachers may become overwhelmed, and so can their computers as they try to process thousands of cases (Rubin & Mokros, 2018; Schultheis & Kjelvik, 2020).

1.2 | From wrangling to pedagogy

Managing the size of datasets is one of the core components of what has come to be known as "data wrangling," (Kandel et al., 2011) the process of cleaning and transforming a dataset in order to prepare it for analysis. While data wrangling is often considered to be a part of professional data practice, it is also often an unexpected requirement of working with educational datasets or designing data-rich classroom activities. In these contexts, educators and curriculum designers have little guidance about how they can wrangle data in ways that do not reduce the benefits of exposing students to "messy" data.

Rubin and Mokros (2018) spoke persuasively about "Goldilocks" data—that is, data that is at a size and level of complexity that engages students without overwhelming or confusing them. If the datasets are too simple, featuring only a few attributes that are related in expected ways, there is no need for exploration, where students try out alternative hypotheses or create different visualizations to reveal a pattern. If the dataset is too small, with too few cases, students may not need statistical or computational thinking in the first place. With only two people from Vermont, for example, you do not need a computer to calculate their median income; more pointedly, that median does not help much if you want to know whether Vermonters in general earn more or less than people from New Hampshire. On the other hand, datasets that are too large or complex are likely to overwhelm students, entice them with spurious findings, or cause frustration as computers lag and crash. What is "just right" depends on the student and the situation, of course; what counts as a conceptually challenging dataset will be different for students in different classes and subjects, or even for the same student later in their course.

Learning to manage the size of datasets is not only an issue for curriculum designers and teachers, but also a skill that students can develop as well as they work their way through school. Already, many public datasets available through scientific organizations are accessed through queries to "data portals" that require similar considerations of which attributes, locations, or time intervals of a given dataset one may wish to access for further exploration (similar data portals are available in CODAP; see "NOAA Weather" and "Microdata Portal" in the Plugins menu). Just as "data moves" (Erickson et al., 2019)—actions associated with preparing data for analysis—reflect important underlying conceptual and technical understandings, querying data can similarly serve as an important site for learning. These skills are related to what the recent GAISE II document calls "considering" data, or determining which attributes are useful and appropriate for a given investigation (Bargagliotti et al., 2020). While our audience of focus for this paper is educators and designers, we hope that it might also offer some insight into whether, how, and why we might introduce data wrangling to students.

1.3 | Datasets: billboard hot 100 and the toxic release inventory

In this paper, our goal is to describe and illustrate a few strategies for reducing datasets to "Goldilocks" size in ways that still help to preserve their assumed benefits. To exemplify these strategies, we will be using two rather large datasets as examples. For simplicity, we call these *BH100* (Billboard Hot 100) and *TRI23CA (Toxic Release Inventory '23 California)*. For each, we link to an example CODAP document, an example Python Jupyter notebook, and the raw CSV (comma separated values) file so readers can engage with the data. The CODAP and Python documents include additional detailed information about how we sourced each dataset. All raw files can be accessed at https://github.com/CalCoRE/how-to-choosy; pre-loaded CODAP links can be accessed at https://bit.ly/bh100-ts and https://bit.ly/tri23ca-ts and pre-loaded interactive notebook links can be accessed at https://bit.ly/bh100-nb and https://bit.ly/tri23ca-nb.

BH100 is a dataset sourced from a GitHub repository (https://github.com/HipsterVizNinja/random-data/tree/main/Music/hot-100) in 2022. It describes every song that charted on the Billboard Hot 100, a US-based industry music ranking, including its highest position on the charts over time. There are over 29,000 cases (songs) describing the weekly positions of the top 100 charting songs per week, each represented by a row in the data table. For each song, there are 13 attributes (such as the song's name, artist, top position on the charts, and musical features such as danceability) represented by columns in the table. The more detailed musical feature attributes are missing for about 5000 of the songs. BH100 was selected to represent a dataset that is "too tall" due to the number of cases in the dataset; it also represents datasets that are "messy" in terms of missing values.

TRI23CA is a dataset sourced from the U.S. Environmental Protection Agency's Toxic Release Inventory, selected to represent a dataset that is "too big" in terms of the number of attributes in the dataset. There are 3482 rows, each representing a reported toxic chemical release event in 2023 in the State of California. For each event there are 119 attributes describing details including information about the releasing facility, location of the event, types and quantities of released chemicals, and the nature of the release (e.g., whether the chemicals were released directly into the environment, treated, transferred to waste management facilities, etc.) represented by columns in the table. TRI23CA was selected to represent a dataset that is "too wide" in terms of the number of attributes in the dataset; these attributes are too numerous to view all at once in a spreadsheet or to simply describe without clustering the attributes into thematic groups.

We selected these datasets to represent topics that are reasonable to explore in school and after-school contexts (music and environmental science, respectively). They each reflect different ways in which a dataset might be introduced. BH100 represents a collection of "interest driven" contexts (e.g., related to youth culture and other topics that are popular among students; Weintrop & Israel-Fishelson, 2024) that are likely to quickly engage students without requiring additional scaffolding about the topic. Datasets like TRI23CA require more preparatory work: students will benefit from exploring news stories, local events, personal experiences, or engaging visualizations to understand the relevance and utility of the dataset for understanding the world around them (e.g., Lanouette et al., 2024; we introduce one such news resource for 23TRICA, below). The datasets also support different types of investigation: for example, BH100 spans several decades which encourages the analysis of trends over time, while TRI23CA includes geospatial data and many categorical variables (we highlight these and other learning goals supported by each dataset in more detail throughout the following sections as strategies are introduced).

Despite all these differences, each dataset shares key features that are often valued by educators: they can support a variety of paths for exploration, and represent contexts that are meaningful and/or consequential to students' lives. The materials accompanying this article include methods for directly accessing these datasets including, where possible, methods for updating the dataset to the most recent available version.

1.4 | Tools: CODAP and Python Jupyter notebooks

In the examples below, we illustrate various strategies for creating "Goldilocks" datasets using two different tools. The first tool is the CODAP, a free online data analysis tool. We created a plugin for CODAP called Choosy, which is designed specifically to get a handle on the problem of too-large educational datasets. Choosy can be found in the Plugins menu of CODAP under "Data Moves" → "Choosy." This tool provides a graphical interface within CODAP (Figure 1) for organizing attributes into groups, and makes it easy to show and hide the attributes individually by switching the visibility of the attribute on or off, or on a whole-group basis by clicking the eye icon next to the group name. It also provides tools for "tagging" collections of cases using new attributes so that those new collections of cases can be filtered, analyzed, and organized. While our focus in this paper is on reducing datasets to a manageable size, Choosy also lets students increase the size and complexity of their dataset when needed, by reviewing and adding hidden attributes over the course of their investigation. In this way, Choosy reproduces some aspects of public data portals, in which users review and identify the set of attributes they are interested in downloading, and may be pedagogically useful in this way as well.

The second tool we use to demonstrate these methods is the *Python Jupyter notebook*, a type of document that

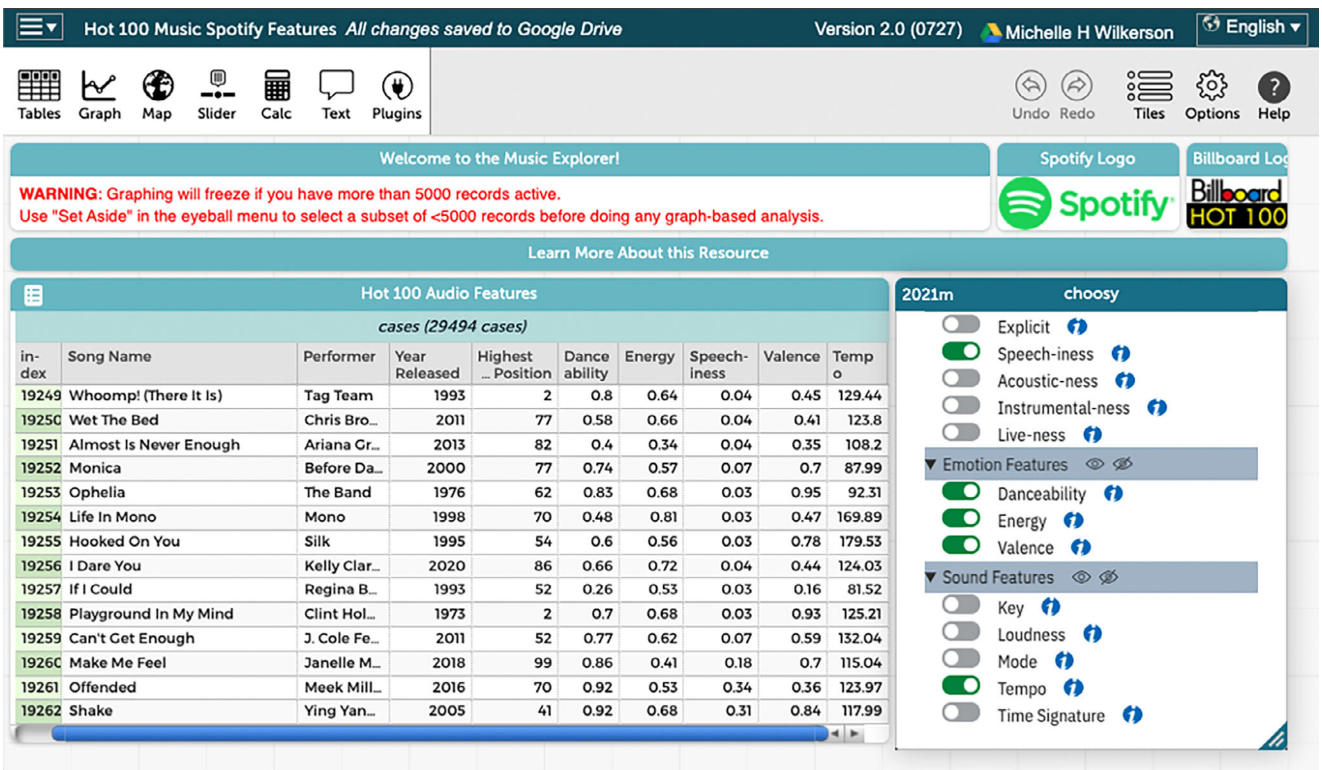

FIGURE 1 The BH100 dataset loaded into our demonstration Common Online Data Analysis Platform (CODAP) document. A CODAP plugin called "Choosy" allows attributes to be grouped and made visible or hidden in the table interface (lower right). [Colour figure can be viewed at wileyonlinelibrary.com]

includes computer code that can be directly edited and run (Figure 2). In this paper, we use these notebooks with pandas, an industry-standard library for structuring, transforming, and analyzing data. In pandas the main object is a DataFrame, which represents a data table as rows and columns, along with a set of special methods to search, access, and perform operations on the table's contents.

We selected these two tools because they are both free and open source, and they are both growing in popularity for teaching data analysis and related skills at the precollegiate level. They also each reflect the different conceptual and technological challenges of working with large datasets. Conceptually, CODAP includes many features designed to help students manage the challenges associated with large datasets, some of which we will illustrate below. But technologically, CODAP is currently only practical with 5000 cases or fewer, especially on types of computers used in public schools. Python Jupyter notebooks, on the other hand, can handle much larger datasets. However, they were designed for use by professional scientists, and require more conceptual scaffolding to examine the content and structure of the big datasets they support.

Of course, there are a number of considerations that educators and designers should be aware of when deciding to use any technology tool as a part of educational activities. One involves technical constraints. While CODAP is a free and web-based tool, users must log in with a Google Account or have file writing privileges on their local computer in order to save their work. Depending on a school's wireless setup, users may encounter bandwidth restrictions if many students are using CODAP large datasets all at once. Similarly, while Python Jupyter notebooks can run on a variety of free-to-use cloud-based online environments, many of these sites may be restricted on school computers, require users to create accounts and login, or otherwise present logistical challenges. And while both tools can be run on tablets, they are best experienced using laptop or desktop

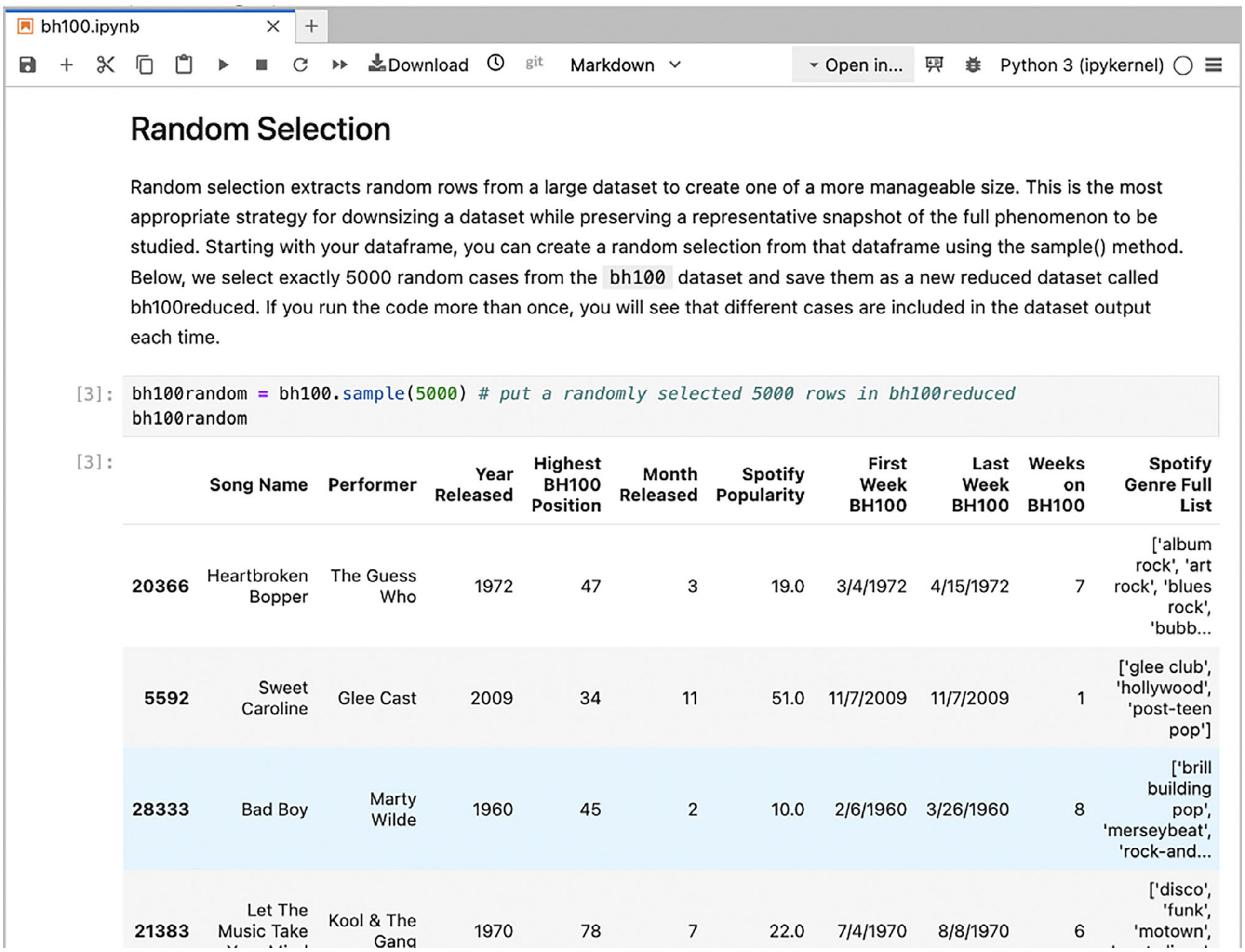

FIGURE 2 The BH100 dataset loaded into our demonstration Python Jupyter notebook and reduced using random selection. We use the pandas library, an industry standard, to simplify the data wrangling process. [Colour figure can be viewed at wileyonlinelibrary.com]

computers with large screens. Teacher preparation is also important. The strategies and demonstrations we present in this article assume that educators are already familiar with loading, visualizing, and analyzing data using CODAP and/or Python Jupyter notebooks. While we have worked to make our supplementary materials as accessible as possible, novice readers may wish to first familiarize themselves with CODAP and Python Jupyter notebook basics. At the time of writing, there are several free and low-cost guides and professional development workshops available online (we recommend Concord Consortium's Getting Started Tutorials for CODAP at https://codap.concord.org/get-started/tutorials/, and Bonaretti's Learn Python with Jupyter text at https://www.learnpythonwithjupyter.com/).

To follow along with the strategies and steps described below, we invite readers to open the example documents that accompany this manuscript. When we describe each strategy, unless stated otherwise, we will assume that a dataset is already loaded into CODAP as a table, or loaded into a Python Jupyter notebook as a named DataFrame using our data abbreviations (BH100 and TRI23CA).

2 | WRANGLING TOO MANY CASES

Let us begin with the problem of too many cases: too many rows in the table. Here, we will demonstrate three different ways to reduce the number of cases in a dataset depending on your learning objectives: *random selection*, *purposeful selection by attributes*, and *building your own selection attribute*. For each strategy, we will identify the main benefits and drawbacks of each approach, describe key statistical and investigative goals that are best supported by the resulting "Goldilocks" dataset, and demonstrate how to actually execute the strategy using CODAP and Python Jupyter notebooks.

2.1 | Random selection

Random selection is exactly what it sounds like—random cases are extracted from a too-big dataset to create one of a more manageable size. This is the most appropriate strategy for downsizing a dataset while preserving a representative snapshot of the full phenomenon to be studied. Another technique that is similar to random selection is *systematic selection*, selecting every *n*th row of a data table. We do not recommend systematic selection unless you have a specific reason for using this method, because it can also lead to unintentionally non-random sampling. For example, if a dataset's cases represent readings taken from sensors A, B, and C, respectively, every minute, removing every ninth record may unintentionally remove only the readings from sensor C. Time series can be an exception to this rule; they sometimes benefit from systematic selection. For example, if you have temperature readings every minute for a year (famously, 525,600 cases), you may be able to answer your question with hourly data (only about 8800). Implementing this in a real-world setting with occasional missing data, however, requires additional care and skill, and belongs in the "build-your-own selection attribute" category described later.

2.1.1 | Pedagogical highlights and drawbacks

Reducing the number of cases in a dataset using random selection is most appropriate when you would like to use a dataset to teach statistical approaches for describing and investigating the general characteristics of the full set of data. A dataset reduced by random selection maintains the representativeness of its parent dataset in terms of the range and distribution of possible values for each attribute. It also preserves the "messiness" of the dataset, in that the resulting smaller dataset will still include missing and erroneous values to the extent that they were included in the full dataset. Because of this, random selection is useful to prepare datasets for activities where students are expected to describe or explore general, aggregate patterns in the data (such as measures of center, distributions of particular attributes, and trends in a time-series dataset).

A major drawback of random selection is that it may remove cases that are of special interest to students, or that represent important instances or outliers that are relevant to the inquiry. We might imagine, for example, that a student working with the BH100 dataset would want to begin by locating their favorite song or artist, or to compare known songs with very high and very low measurements on a given indicator. Focusing on a particular case can help the student to contextualize the data, helping them understand how abstracted attributes such as "speechiness" or "danceability" correspond to the features of songs they know well. This, in turn, may inspire more statistical questions (e.g., "Are songs released in the last 5 years more 'danceable' than songs in the past?"). It could be discouraging or confusing for the student not to find a song that they know to have been very popular. Random selection also reduces the number of cases that represent any meaningful "slice" of the data (e.g., in certain genres, by certain artists, and during certain eras), making comparisons of subgroups and other deeper investigations that benefit from higher data resolution

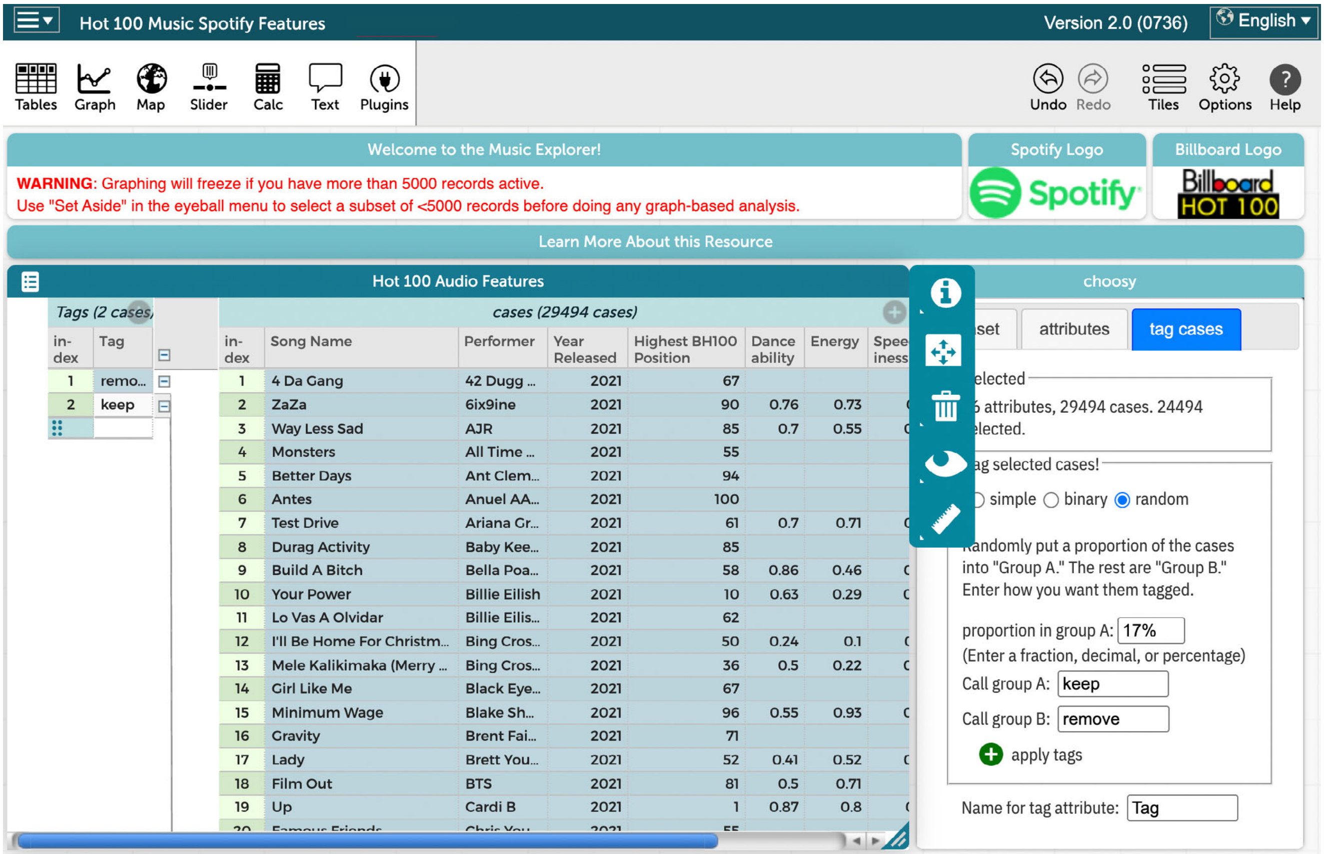

FIGURE 3 The Choosy "tag cases" tab is used to create an attribute that assigns 17% of BH100 cases to a "keep" group. [Colour figure can be viewed at wileyonlinelibrary.com]

less appropriate (*Stratified* sampling can address this problem in some circumstances, but we will not address that technique here).

2.1.2 | Getting it done

In CODAP, random selection can be accomplished in one of two ways. When a dataset with more than 5000 cases is first imported into CODAP, the user is automatically asked if they would like to load a random sample of a smaller size (5000 cases by default) or to select every *n*th record. Alternatively, if a too-big dataset has already been loaded into CODAP, you can use random selection in the Choosy plugin to reduce the dataset using the "tag cases" tab (Figure 3). Select the "random" option, and indicate what proportion of the cases you would like to keep (for our 29,000 BH100 cases, we would elect to keep about 5000/29,000 = 17%). Use the textbox "Call group A;" to mark the selected group of 17% of cases "keep," and use "Call group B:" to mark the other group of remaining cases "remove" (you can call these whatever you did like), and press "apply tags." This will create a new attribute in the dataset titled "Tag" (you can change the name of the attribute using the "Name for tag attribute:" textbox at the bottom of this tab).

Now, you can select and delete all of the cases called "remove," keeping your reduced dataset of (about) 5000 randomly selected cases. The easiest way to do this is by dragging the new "Tag" column that appears in the table to the far left, to form groups. Click on the "remove" group to select all corresponding cases (this may take a while), and then use the trashcan icon to remove all other cases by choosing "Delete Selected Cases." Note that in a different teaching context—suppose you are doing a lesson in machine learning (see, for example, the work of Podworny et al., 2025)—you can use this exact same technique to randomly label a set of cases as "training" or "test."

With Python pandas, you can create a random selection from a too-big dataset using the sample() method. To select exactly 5000 random cases from the BH100 dataset and save them as a new reduced dataset called bh100reduced, run the following command:

$$\text{bh100reduced} = \text{bh100.sample}(5000).$$

2.2 | Purposeful selection by attribute(s)

Purposeful selection involves reducing a dataset so that it only includes cases with certain characteristics related to one or more attributes. This method is appropriate if you suspect that many cases in your too-large dataset are not useful or usable for your intended activity. For example, this method allows you to remove cases if they are missing data about an important attribute you intend to focus on in your lesson or analysis. Purposeful selection is also useful if you want students to limit their investigation to some subset of the dataset that satisfies certain conditions (e.g, keeping only cases within certain categories, geographic regions, time periods, or that otherwise fall within a restricted range of values). In fact, we generated our second dataset (TRI23CA) by purposefully selecting only cases representing events in California from a national TRI 2023 dataset.

2.2.1 | Pedagogical highlights and drawbacks

Purposeful selection of subsets of data by attribute can be especially useful when building activities or pursuing projects that involve making comparisons across groups in a dataset. For example, we can create a dataset that includes only songs from the first and last years available in the dataset to compare the acoustic characteristics of these two groups. When used carefully, this technique can also familiarize students with common issues when using "messy" datasets, such as unequal group size and sampling bias.

Whereas random selection preserves the variability of a full dataset, purposeful selection intentionally selects for certain characteristics that will likely result in a non-representative subsample of data. This means that students working with purposefully selected datasets should understand that they do not have enough information to make general claims about the broader dataset or the world it represents. Instead, the strength of purposeful selection is that it creates datasets that are of manageable size, while still allowing for "deeper dives" that require greater data resolution such as making comparisons between certain subgroups or exploring a particular situational state or condition represented within the data in detail.

Purposeful selection can also be used to introduce student-driven and collaborative features to your educational plans. For example, different sub-selections of the same dataset can be used to have different student groups examine and describe the key characteristics of songs from each decade of the BH100 dataset (1960s, 1970s, and so on), and then compare the decades as a class to make claims about the ways popular music has changed over time. Purposeful selection is also well suited for student-driven investigations: as they pose investigative questions, students themselves might identify particular artists, musical genres, song characteristics that can be used to wrangle a large dataset into something more manageable. For example, a student might ask the investigative question, "What proportion of BH100 songs 'one-hit wonders' (in which only one song by a given artist is ever listed on the charts)?"

At the same time, purposeful selection brings complicated issues concerning sampling bias and inference. For example, imagine a dataset that is purposefully selected from BH100 to include only songs that have reached #1 at some point in their time on the charts. The mean number of weeks that the songs in this purposefully selected dataset have remained on the charts in any position for this subset of songs is much higher than the mean number of weeks that *any* songs that have remained on the charts. It makes sense that songs that have reached #1 are more likely to spend prolonged periods of time on the charts in any position, but it is also easy to imagine students might use the dataset of only #1 songs to develop claims or expectations about the behavior of the Billboard charts more generally, if they are not careful.

2.2.2 | Getting it done

To illustrate purposeful sampling of datasets by attribute, let us imagine that we want to explore or ask questions about the musical features of only the songs that have reached #1 on the Billboard chart. Since the dataset includes an entry for each song that placed in *any* position on the Billboard Hot 100, most cases are not relevant to our question.

In CODAP, let us first remove all the cases where a song was not in first place on the charts. There are many ways to do this, but our favorite way is to drag the "Highest BH Position" attribute to the left in the table to create a hierarchical group. We can then use the same method we used above to select only the group of cases where the highest position is 1, and remove all other cases by choosing "Delete Unselected Cases" (Figure 4). This leaves 1133 cases representing only songs that have topped the Billboard charts.

In Python pandas, we follow a similar process using dataframe selection methods. To get all the cases in a dataframe that meet certain conditions, use the expression dataframe[*condition*]. For us, we want cases where the attribute in BH100 called "Highest BH100 Position" is equal to 1, so our *condition* is: bh100['Highest BH100

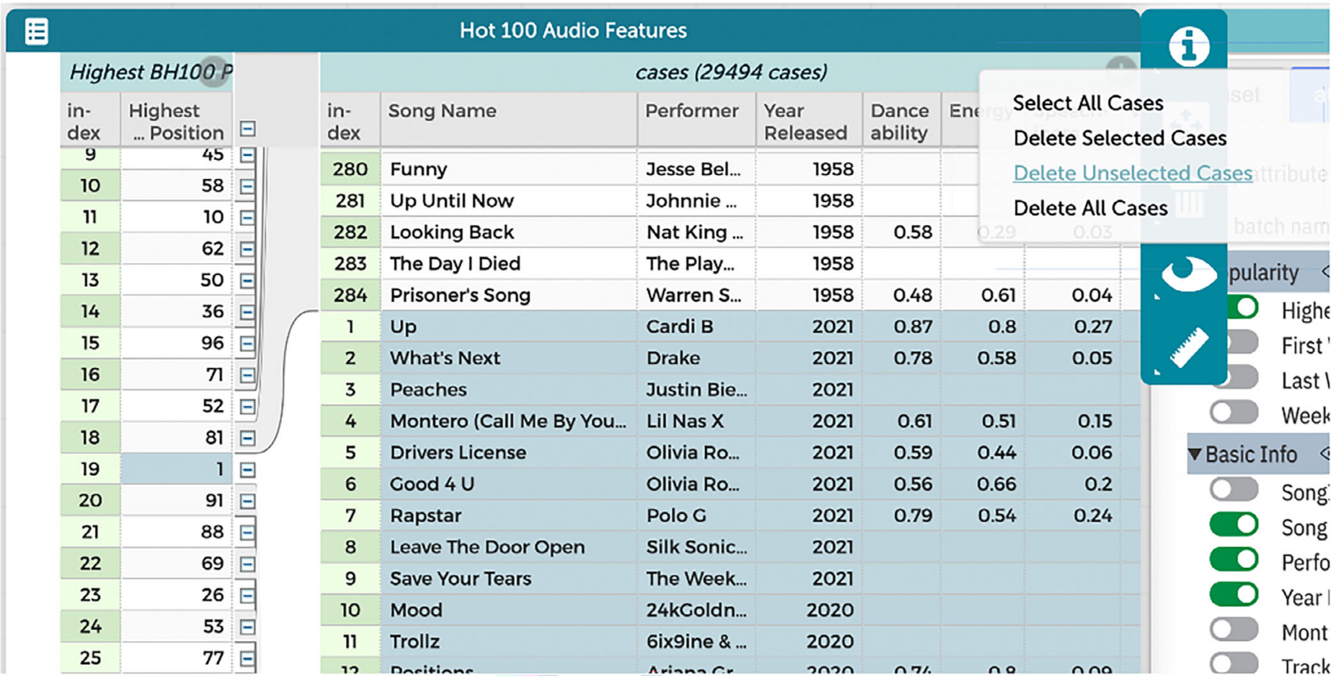

FIGURE 4 Using hierarchy, case selection, and "Delete Unselected Cases" in Common Online Data Analysis Platform to reduce dataset size. [Colour figure can be viewed at wileyonlinelibrary.com]

Position']==1 (in plain English, "cases in bh100 where the value of 'Highest BH100 Position' is equal to 1"). Therefore, to access all of the cases in BH100 that meet this condition, our final expression is bh100[bh100['Highest BH100 Position']==1]. To save this new reduced dataframe, we use bh100reduced = bh100[bh100['Highest BH100 Position']==1].

2.3 | Build-your-own selection attribute

There are other ways of creating a smaller dataset based on information that is not already available in the dataset itself. These could be specific cases that you identify manually, or cases you might identify using computer code to extract some new, meaningful information from the attributes you already have. We will call this a "build-your-own" dataset. For example, when first becoming familiar with the BH100 dataset, a teacher may wish to start with a small dataset including only students' favorite songs. Or, there may be particular features or combinations of attributes that can be used to filter cases (such as the presence of particular words in a song title). Build-your-own selection attributes can also be useful for interdisciplinary lessons. For example, an educator may wish to collect a small set of news releases about various chemical spills, and allow students to examine those spills in particular within the TRI23CA dataset before examining broader patterns.

2.3.1 | Pedagogical highlights and drawbacks

Using a build-your-own selection attribute offers considerable flexibility in crafting a dataset that meets your instructional needs. For example, an educator may use data with particular attributes to illustrate the effects of selection bias, by highlighting how descriptions of data with one shared attribute may be less representative of the full dataset in other ways as well. They can also use build-your-own selection attributes to reduce a dataset's size in a way that allows students to examine more relevant or meaningful subsets with high resolution. Build-your-own datasets can also help educators and designers craft thematically interesting investigations through the strategic selection attribute characteristics.

Just as with purposeful selection by attribute, creating your own selection criteria does not guarantee that the representative characteristics of the full dataset will be preserved in the smaller resulting set. It allows teachers and students the most customization to make datasets "their own," but it can take considerable time and planning to select and identify which cases should be included and to consider how those decisions will shape what analyses and claims are appropriate. Build-your-own datasets can also present the problem of too much choice—rather than learning to construct questions that can be answered by a particular dataset, students may instead try to "force" a dataset to relate to less appropriate questions or to focus on specific cases of interest rather than on broader patterns. Because the criteria for including a record in a dataset can be almost anything, it is best to frame activities using build-your-own datasets as *descriptive*—getting to know the features of a particular set of cases, rather than using those cases to make general claims or comparisons. Or, as with purposeful selection using existing attributes, it might be presented as *comparative*, with an expectation that the selected subset of data is likely to differ from the full dataset in meaningful ways.

Build-your-own selection attributes can also allow for another dataset size reduction strategy, *aggregation*. We will not dwell on this technique here because it is so different, but it is an important one for teachers and designers to consider. Especially with truly huge datasets, it may be possible to group or "bin" the original, atomic microdata into a large number of subsets, and calculate new attributes (e.g., means, proportions, or initial values) for those groups based on the underlying attributes in the microdata. This set of data moves—grouping and summarizing—by definition reduces the number of cases, but *changes the nature of a case*. Using the TRI23CA dataset, we could group toxic events by *county* and study those, rather than study the individual events.

2.3.2 | Getting it done

The easiest way to create a build-your-own dataset using your own selection attribute in CODAP is manually. Hold down the Command key on your keyboard and click to multi-select all the cases you did like to keep in the dataset. Then, use the Choosy "tag cases" tab, select "simple," enter a name (like "keep") for the selected cases as the Tag value, and click "tag selection." This will create a new attribute called "Tag" in the dataset, where all of the cases you had selected will be assigned the indicated value. You can also do something similar without Choosy. Use the plus icon in the upper right-hand corner of the data table to create a new attribute. You can double-click inside each cell of the column corresponding to this attribute for each case to indicate with a symbol ("Y," 1, or "keep," etc.) which cases to keep and which to remove. For example, you might invite each student to identify two to three song titles they did like to add to the classroom's collective build-your-own dataset. You can mark these songs in the new attribute (let's assume it is titled "Favorites"), and then use the method described in Section 2.2 above to keep only the selected songs.

For some types of inquiry, however, it is not practical to review all of the cases by hand. Let us imagine a student wishes to explore only the songs from the dataset that have the word "love" in the title (e.g., "Have love songs become 'speechier' over time?"; or "What musical features are most prevalent in songs that have 'love' in the title?"). For these kinds of more complex custom queries, CODAP includes tools to build formulas, which we can use to generate a new selection attribute for building our "Goldilocks" dataset. First, we create a new column in our data table representing an attribute called "has-love." We can then right-click on that attribute heading and select "Edit Formula …" CODAP includes many functions that allow text to be searched and manipulated. One function called patternMatches counts the number of times a specified text pattern appears in a given attribute. In Figure 5, we use patternMatches('Song Name', "love") to check and see if the word appears in the song title for each case in the dataset. This function makes the value of the "has-love" attribute equal the number of times the word "love" appears in the song title. Now, you can click on the attribute name "has-love" and select "Delete Formula (Keeping Values)" to make this column a standard part of the dataset, rather than a function. Using the same method that was described in Section 2.2 above, you can now delete cases with 0 in the "has-love" attribute to generate a dataset with only the 2492 cases that have the word "love" in the Song Name attribute at least once.

With Python pandas, the process is a bit different. When constructing build-your-own datasets manually, users do not have easy access to scroll through cases to identify favorites. Instead, you need to find out which rows of the table have the songs you want, and then add them one-by-one to create a reduced dataset. We do this in our example notebook by searching for records that contain our desired song names. We find one for which we know a full name using bh100[bh100['Song Name'] == "Skyfall"], and we perform a more general search for more songs using bh100[bh100['Year'] == "1995"] to find the indices of other desired songs (you can also search other attributes, such as 'Performer'). Once we have found the index of each song to include, we store them in a list called myIndices. You can then create a data table that only includes the cases with indices in the list using bh100reduced = pd.DataFrame([bh100.loc[i] for i in myIndices]).

While CODAP is better suited for reviewing and selecting cases by hand or using simple queries, pandas shines when it comes to more systematically selecting cases based on more precise and complex queries, including ones that can look for multiple words or particular capitalizations. We make a new table that only has cases with "love," "Love," or "LOVE" in the song title using a function called contains: bh100reduced = bh100[bh100['Song Name'].str.contains("Love|love|LOVE")]. This

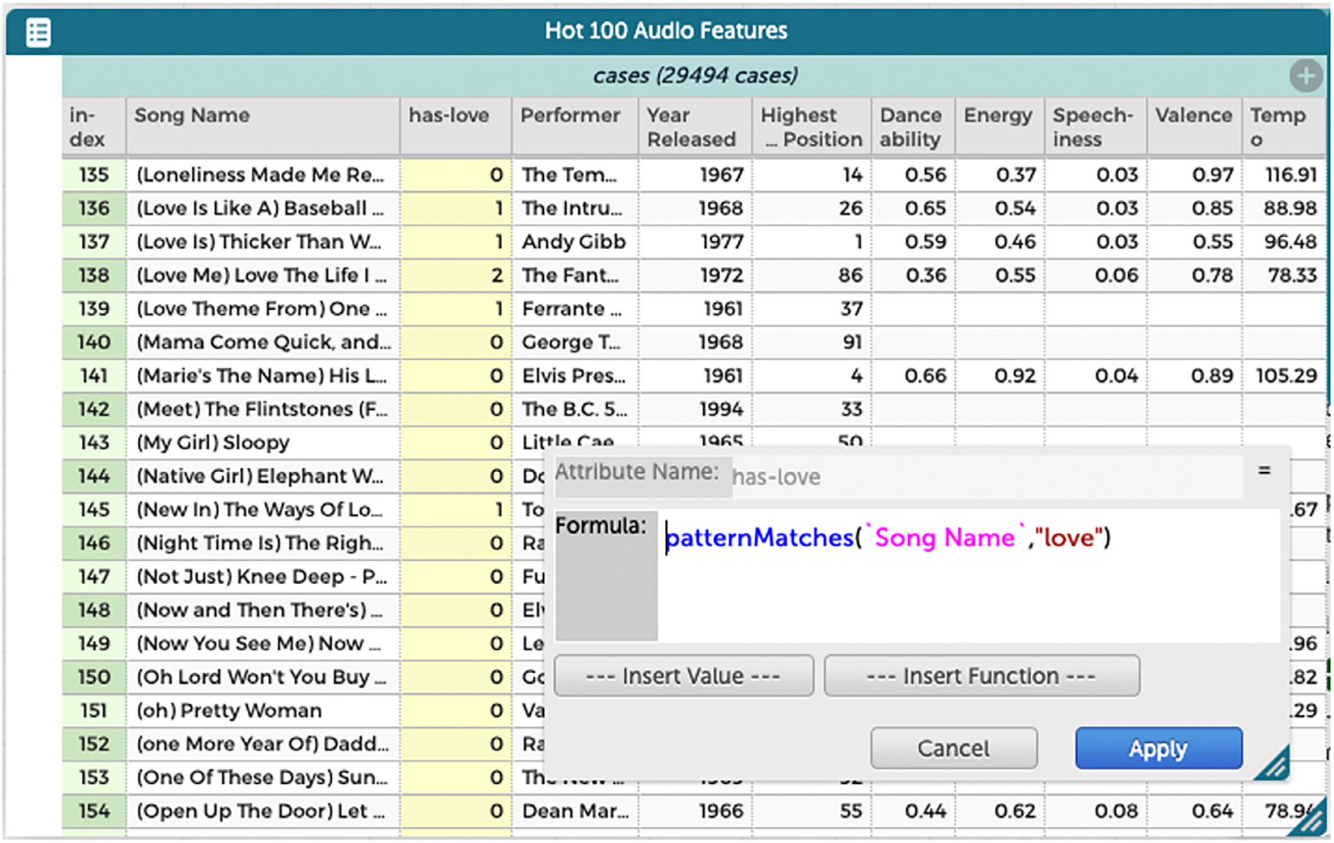

FIGURE 5 Using Common Online Data Analysis Platform functions to create an attribute that identifies songs with "love" in the title. [Colour figure can be viewed at wileyonlinelibrary.com]

produces a data frame with the same 2492 cases identified above using CODAP.

3 | WRANGLING TOO MANY ATTRIBUTES

Another common issue when using large public datasets for educational purposes is the problem of too many attributes. Environmental datasets such as TRI23CA can include hundreds of specific indicators; survey datasets from organizations such as the Pew Research Center similarly report scores of questions per participant. While having access to so many attributes can enable the pursuit of a variety of investigative questions, it can easily become overwhelming. Working with these datasets requires planning and thoughtfulness to consider: which attributes are *actually* connected to my research question? How were each of these attributes measured, and what might be the limitations or biases related to these attributes in the dataset? How do I prevent myself from simply digging for what may turn out to be spurious relationships in so much data?

In CODAP, the Choosy plugin lets us slowly introduce students to datasets in a way that is structured, yet flexible. You may have already noticed in the screenshots and CODAP document associated with the BH100 dataset that its attributes were organized into smaller, thematic groupings such as "Basic Info," "Popularity," and "Sound Features" using the Choosy plugin. Similarly, we have grouped attributes in the TRI23CA dataset into different collections of information about each toxic release event (e.g., "Facility," "Location," "Toxic Release Basics," "On Site Release"). Thinking of attributes as something you can hide or display according to circumstances offers a variety of ways to simplify and focus students' work with a dataset.

For example, when you first open the TRI23CA dataset in CODAP, you will see only a few attributes displayed: the facility name, city, and industry sector, along with basic information about the toxic release such as whether the toxins released are identified in the Clean Air Act, or whether they are known to be carcinogenic, bioaccumulative, or a per- or poly-fluoralkyl substance (also known as "forever chemicals"). This initial set of attributes is designed to get students thinking about why toxic release events are relevant to their lives, brainstorming what questions they might be able to explore using this dataset, and considering what other information they wish they had about these events. This grouping function can be reproduced in Python Jupyter notebooks by compiling lists of attribute names, and using those lists to more intentionally select and display meaningful subsets of attributes in the data frame.

To further help scaffold students into working with these types of datasets, we propose three strategies below for reducing and organizing too many attributes: *thematic selection*, *mathematics-driven selection*, and *question-driven-selection*. Just as we did for too many cases, for each attribute management strategy we will identify benefits and drawbacks, key learning goals, and provide practice demonstrations using the CODAP and Python Jupyter notebooks tools. We will exemplify these strategies with a focus on the TRI23CA dataset, which has a total of 119 attributes.

3.1 | Thematic selection

Thematic selection involves splitting a dataset up into groups of conceptually related attributes, often that measure or describe the same underlying system component. For example, several attribute groups in TRI23CA aggregate information about various methods of toxic release—on site releases, off site releases, off site recycling, and so forth. These groups are meant to help students brainstorm appropriate questions and examine what different indicators are available to help them address those questions. For example, depending on the particular types of chemical toxins students might be interested in, they can then identify the mode(s) in which those toxins are introduced into the environment (e.g., through release into the air, water, or underground) by honing in on attributes collected within the appropriate thematic group, such as the "On Site Release" (Figure 6).

Thematic selection lets you group many attributes together under broader themes, and then easily hide or show them *all* as needed with a single click. It can also make explicit the larger conceptual structure of an overwhelmingly large dataset for students. For example, the BH100 dataset includes groups that emphasize basic song information, information about a song's popularity, genre information, performance information, information about the song's emotional cadence, and so on. We created these groups to highlight potential relationships students may wish to pursue in more detail: for instance, they may want to explore whether certain musical characteristics correspond to particular genres or emotional valences of songs.

3.1.1 | Pedagogical highlights and drawbacks

Thematic groups can be useful when you want to orient students to the underlying hypotheses and relationships

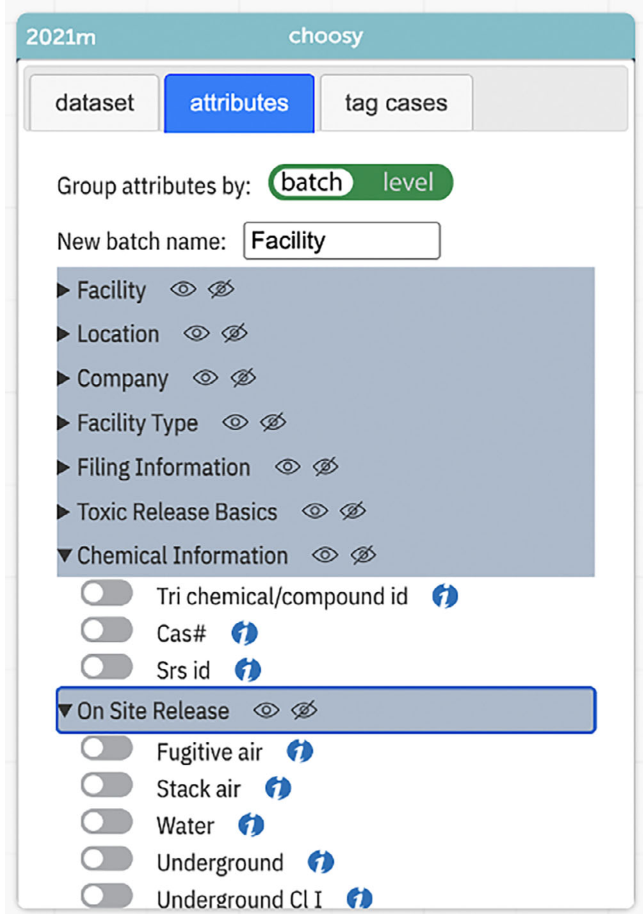

FIGURE 6 The Common Online Data Analysis Platform Choosy interface, with thematic groups from the TRI23CA dataset. [Colour figure can be viewed at wileyonlinelibrary.com]

that will motivate an investigation. For example, you might ask students to identify a thematic group and justify why they think it is important to investigate, or to make predictions about what they will learn from the attributes in that cluster, before they get started with their analysis. Thematic selection encourages students to think about what questions they might actually be able to answer with a given dataset, and which of the available attributes in a dataset are most relevant to their inquiry. This grouping can help limit students to a small number of attributes that are most likely to relate to their inquiry, reducing "fishing expeditions" and making it easier for educators to structure systematic activities in which students describe, hypothesize about, and compare attributes in a dataset. And, because related attributes are grouped together, thematic selection can also support considerations of confounding variables.

Thematic selection can be especially useful for jigsaw-like activities, in which different groups explore different aspects of an interconnected system. For example, some student groups exploring the TRI23CA dataset might decide to focus on toxin releases at facility sites across the state (e.g., via air, water, underground, or other methods; this is the "On Site Release" attribute cluster), while others may focus more broadly on how facilities handle toxic chemicals (e.g., released on site, transported off site, treated, recycled, etc.; this is the "Summaries" attribute cluster). Student groups may then put their results together—which will require them to practice describing nested proportions—to address more complex questions such as "What reportable toxins are most likely to be released on site?" Thematic grouping can be used to encourage students to explore relationships between particular components of a system, and it can make it easier for students to locate the attributes they need to address research questions of their choosing.

A downside of thematic selection is that there are still many options for students to investigate, and communicating the logic behind the "themes" (groups) selected and what they might reveal is not always easy. Depending on how thematic groups are determined, it can be likely that attributes within a given group are confounded, and it is important to support students in distinguishing ideas such as correlation, variability, and causality as relationships are identified. Special care should be taken when thematic grouping is used to explore multivariate relationships. While hiding or grouping attributes can simplify a dataset and make it less overwhelming for students, students may still need additional support to develop appropriate research questions or to know which specific attributes will help them answer those questions.

3.1.2 | Getting it done

In CODAP, you can use Choosy for thematic selection using the "attributes" tab. To create a new group based on a theme, type a group name into the "New batch name" textbox in Choosy and press enter. The group will appear at the bottom of the list of attributes below. To add attributes to a group, drag the attribute's name in the list to the group name, making the group's title bar yellow (see Figure 7).

In Python, thematic selection can be done by defining lists of attribute names that should be grouped together. For example, to reproduce the grouping of attributes called "location" that can be seen in the Choosy window of Figure 7 above, we can define a list of attributes in Python using location = ['Street address,' 'City,' 'County,' 'St', ...]. Then, a reduced dataset that only includes selected attributes can be created by specifying the desired thematic lists in brackets. For example, to create a dataset that only includes the location and basic release information lists that are created in the sample notebook for each toxic release event, use tri23ca[location +basicInfo].

3.2 | Mathematics-driven selection

Similar to thematic selection, mathematics-driven selection allows educators and students to focus only on certain attributes that are known to align with their pedagogical goals. But whereas thematic selection focuses on the real-world contextual features of a system, mathematical selection focuses on quantitative features of attributes in the dataset such as their data type, distribution, or mathematical associations with other attributes. For example, a dataset might be grouped into attributes that are known to be positively or negatively correlated with one another—imagine a group of health indicators, a group of attributes that contribute positively to health such as exercise or eating vegetables, and another group of attributes that contribute negatively to health such as smoking or limited healthcare access. If students are encouraged to select attributes from these different groups to compare, this could motivate discussions of direct and inverse relationships. At more advanced levels, it could help to lay the conceptual building blocks for comparison and discussion of R^2 as a measure of the amount of variance in one attribute as explained by others. In this way, mathematics-driven selection can provide a structure to direct students' attention to certain kinds of mathematical features and relationships.

3.2.1 | Pedagogical highlights and drawbacks

Mathematics-driven selection is most appropriate when an educator or designer would like to highlight or take pedagogical advantage of the technical features of a dataset. An educator interested in exploring the impact of missing values on analysis with their students might create a group that features only attributes that have many missing values (in TRI23CA, this includes the "One-time release" and "Bia" attributes). An activity could then ask students to select and examine one or more attributes within that group. By offering groups of attributes with similar mathematical characteristics, this design balances the flexibility of choice that "messy" datasets provide with structured attention to particular data concepts.

In our demonstration datasets, some of the existing groups in the TRI23CA can be considered to be mathematics-driven in that they group attributes of the same data type: the "Basic Information" group includes a

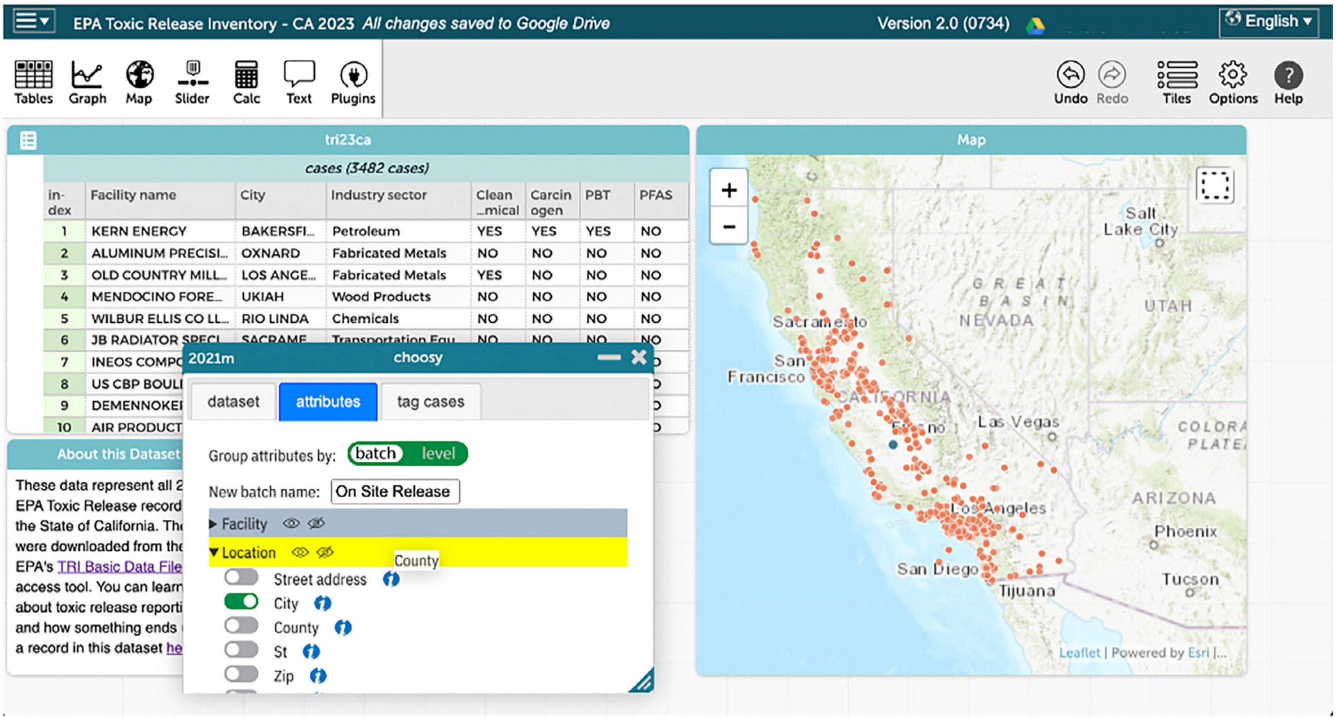

FIGURE 7 Using the Choosy plugin to organize attributes. The user is dragging the "County" attribute to add it to the "Location" group, which is highlighted in yellow. [Colour figure can be viewed at wileyonlinelibrary.com]

variety of important Boolean attributes specifying whether certain risk factors are present for a given chemical release. Other groups like "On Site Release," "Off Site Release," and "Off Site Recycling" include attributes that specify the amount of chemical that was released in a given modality as a continuous quantity. Focusing student attention using groups like these can help structure students' exposure to and practice with different data types, including univariate analysis (e.g., Boolean attributes in the "Toxic Release Basics" group; categorical and geographic attributes in the "Location" group, or continuous quantitative attributes in the "On Site Release" group) and different forms of bivariate analysis (e.g., by having students explore relationships between categorical attributes, a quantitative vs. a categorical attribute, and so on). This, in turn, can help students recognize the sorts of questions they may ask of different data types (e.g., "What proportion of toxic release events involve on site release?," or "Which type(s) of on site release are associated with higher volumes of released toxins?").

Mathematics-driven selection should be used with caution, and carefully framed for students. It should not be positioned as a way to explore or make causal inferences, but rather as a way for students to consider how attributes might be more complexly related, including through confounded relationships. Mathematics-driven selection may be especially useful for highlighting quantitative differences between models, such as between exponential models with different rates, and qualitative differences between model types, such as between exponential vs. linear models.

3.2.2 | Getting it done

In both CODAP and Python Jupyter notebooks, the method for creating mathematics-driven groups is logistically equivalent to the method for creating thematic groups. Simply assign attributes to the appropriate group by attending to the desired mathematical properties you wish to emphasize, rather than their thematic or conceptual associations. Conceptually, however, the method for creating mathematics-driven groups can be quite different from the method for creating thematic groups. This is because, especially in datasets with very many attributes such as TRI23CA, getting a handle on the different mathematical patterns that are available to explore can be difficult in itself.

CODAP allows for simple exploratory review of attributes through visualization. In Figure 8, we demonstrate how relationships between some Boolean attributes of TRI23CA can be quickly examined by selecting the "Graph" menu item and dragging and dropping attribute names onto the graph's axes. Axes can also be clicked to open a menu to select attributes to plot.

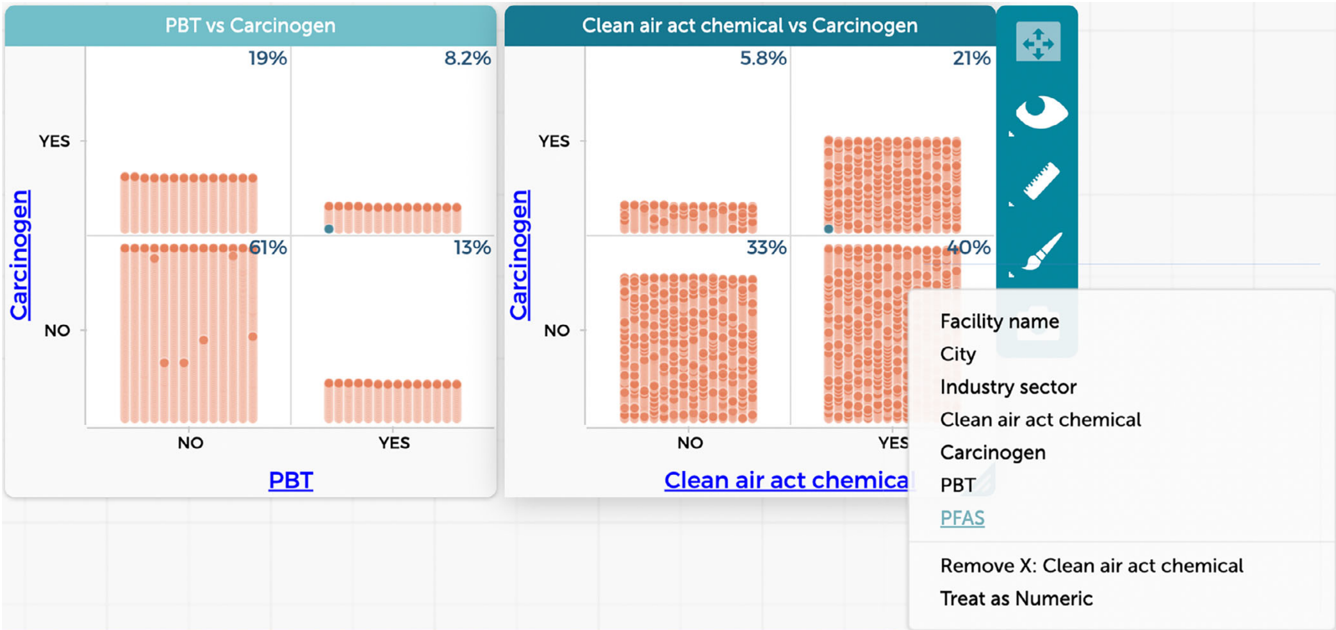

FIGURE 8 Drag-and-drop plotting allows for quick exploration of relationships between attributes when planning mathematics-driven selection. [Colour figure can be viewed at wileyonlinelibrary.com]

Python Jupyter notebooks include some built-in facilities to quickly summarize multiple attributes, and to investigate relationships between attributes. In our example notebooks, we use the .corr() function with a set of quantitative attributes representing the energy, tempo, and other acoustic features of songs in the BH100 dataset to quickly visualize pairwise correlations in this group. This reveals a range of correlations between these attributes (ranging from −0.14 to 0.68). This variety of correlational relationships can provide an engaging context for students to visualize, practice describing, comparing, and reasoning about the strengths and directions of different correlations. Similarly, in the TRI23CA sample notebook, we use a simple loop and the pandas. crosstab() function to examine relationships between Boolean attributes. Strategies like these can help educators and designers more quickly identify focal mathematical patterns and attribute combinations that can best support particular pedagogical goals.

3.3 | Question-driven selection

Finally, educators or designers may wish to identify and group attributes that are most appropriate to address a certain question. Such questions may be posed by students, teachers, or even within curriculum materials themselves. For example, different students examining the TRI23CA dataset may wonder about different aspects of this phenomenon—some may be curious about the geographic distribution of reported events; others about the risk factors posed by events; and still others may wish to investigate releases of particular chemicals or classes of chemicals. In our work, we have used Choosy to build activities in which students form groups based on the different questions they are interested in. Each group then receives a small set of 3–5 "shared" attributes that all students can view by default (in the TRI23CA dataset, this might be basic information such as the name, industry, and date for each reported toxic release event), along with a subset of additional attributes that are most appropriate to address their selected question (e.g., location information, chemical risk classifications, or more specific chemical information, respectively, to address the curiosities posed above).

3.3.1 | Pedagogical highlights and drawbacks

Question-driven selection is the most clearly student-centered approach to wrangling datasets, which can promote engagement and a sense of agency among students. However, for question-driven selection to work well pedagogically, students must have an understanding of the data context. Educators may wish to kick off question-driven activities by providing students with news articles or videos about the system the data represent. One such context-builder for the TRI23CA is the 2023 article series *Hidden Hazards*, published by CalMatters, which

TABLE 1 Summary of data reduction strategies, pedagogical highlights and drawbacks, learning goals, and standards connections.

Case (row) reduction strategy	Highlights	Drawbacks	Learning goals supported	Example GAISE II and IDSSP connections
Random selection	Retains aggregate characteristics of the full dataset.	Specific cases of interest to students, including outliers or cases that provoke deeper questions/ connections, may be removed.	Understanding descriptives: distribution, trend, center. Exploring mapping from sample to population. Exploring training/ testing sets.	*GAISE II* 1. Formulate statistical investigative questions a. Pose summary, comparative, and association statistical investigative questions about a broader population using samples taken from the population. 2. Analyze the data a. Describe key features of distributions (center, variability, and shape). b. Represent the variability of categorical and quantitative variables using appropriate displays. *IDSSP* 1. Basic tools for exploration and analysis a. Single variable, pairs of variables, three or more variables. 2. Graphs and tables.
Purposeful selection by attribute(s)	Focuses on cases students know more about, or that will address specific questions.	Cases associated with the selected attribute may not be representative of larger patterns in the data. Sampling bias.	Describing and comparing different populations. Collaborative "jigsaw" activities that require clear communication of findings. Hypothesis testing across groups.	*GAISE II* 1. Analyze the data a. Use reasoning about distributions to compare two groups based on quantitative variables. b. Explore patterns of association between two quantitative or categorical variables. 2. Interpret results a. Describe the difference between two groups with different conditions. b. State the limitations of sample information. *IDSSP* 1. Basic tools for exploration and analysis a. Pairs of variables (numeric × categorical; numeric × numeric). 2. Avoiding being misled by data a. Discuss examples of selection biases. b. Discuss random sampling as a strategy for overcoming selection biases.

TABLE 1 (Continued)

Case (row) reduction strategy

	Highlights	**Drawbacks**	**Learning goals supported**	**Example GAISE II and IDSSP connections**
Build-your-own selection attribute	Focuses on issues defined by the teacher or students. Can motivate "data moves."	Difficult to assess representativeness or bias in the resulting dataset. "Forced" investigations that do not align well with data. Too much choice.	Data cleaning, data wrangling, data "moves." Practicing descriptive statistics. Describing and comparing populations.	*GAISE II* 1. Formulate statistical investigative questions a. Pose statistical investigative questions for data collected from online sources. 2. Consider data a. Interrogate the data set to understand the context of the variables as they may relate to statistical investigative questions. b. Use technology to subset and filter data sets and transform variables. *IDSSP* 1. The data-handling pipeline a. Recognize situations where simple data cleaning and transformation is important. b. Implement simple data cleaning and transformation operations.

Attribute (column) reduction strategy

	Benefits	**Drawbacks**	**Learning goals supported**	**Example GAISE II and IDSSP connections**
Thematic selection	Emphasize modeling, theory-driven analysis. Makes explicit the conceptual structure of the domain of study. Highlight data context; specific relationships of interest.	Grouped attributes may be interpreted as simply illustrating a target relationship as a goal, rather than to support inquiry. Emphasis on theory may encourage focus on spurious relationships.	Exploration of relationship between theory and data (e.g., in science investigations where the mechanism is known). Presenting and communicating findings.	*GAISE II* 1. Consider data a. Interrogate the data set to understand the context of the variables as they may relate to statistical investigative questions. b. Understand that data are information collected and recorded with a purpose and can be organized and stored in a variety of structures. 2. Analyze the data a. Summarize and describe relationships among multiple variables. 3. Interpret results a. Use multivariate thinking to understand how variables impact one another.

(Continues)

TABLE 1 (Continued)

Attribute (column) reduction strategy	Benefits	Drawbacks	Learning goals supported	Example GAISE II and IDSSP connections
				IDSSP 1. Basic tools for exploration and analysis a. Pairs of variables; three or more variables.
Mathematics-driven selection	Emphasizes specific mathematical content and skills. Useful for introducing students to specific data types, plots, models.	Attributes may be interpreted as simply illustrating a target mathematical pattern as a goal, rather than to support inquiry. Emphasis on mathematical features of the dataset may reduce focus on data context.	Understanding data types and their corresponding methods of analysis, visualization, and modeling. Exploring and describing details of specific patterns/models (e.g., different distributions, strengths/directions of relationships).	*GAISE II* 1. Analyze the data a. Learn to use the key features of distributions for quantitative variables. b. Explore patterns of association between two quantitative variables or two categorical variables. c. Identify appropriate ways to summarize quantitative or categorical data. 2. Interpret results a. Acknowledge the presence of missing values and understand how missing values may add bias to an analysis. b. Use multivariate thinking to understand how variables impact one another. c. Communicate statistical reasoning and results to others in a variety of formats. *IDSSP* 1. Basic tools for exploration and analysis a. Single variable, pairs of variables, three or more variables.
Question-driven selection	Emphasizes full data investigation cycle. Allows for student-centered explorations. Sets a context for prediction and hypothesis generation; prioritizes (statistical) depth.	Students require support to pose appropriate statistical investigative questions. Difficult to anticipate the skills and models needed to pursue student-posed questions.	Completing the full data investigative cycle. Evaluation and communication of data analysis results. Hypothesis testing.	*GAISE II* 1. Formulate statistical investigative questions a. Pose statistical investigative questions for data collected from online sources. 2. Collect/consider data a. Understand that data are information; recognize that to answer a statistical investigative question, a person may [...] use data that have

TABLE 1 (Continued)

Attribute (column) reduction strategy				
	Benefits	Drawbacks	Learning goals supported	Example GAISE II and IDSSP connections
				been collected by other people for another purpose. b. Interrogate the data set to understand the context of the variables as they may relate to statistical investigative questions. *IDSSP* 1. The data-handling pipeline a. Exploring/analyzing the data.

investigates what happens to toxic waste in the state of California (CalMatters, 2023). We have found it useful to present such materials before students access a dataset, and then to encourage them to develop initial questions and brainstorm what kind of data they would need to address those questions.

Question-driven selection also requires students to have a solid foundation in posing investigative statistical questions, once they are introduced to a dataset. These are questions that are answerable through available data, that anticipate variability in the data that will be used to answer the question, and that precisely refer to the attributes and populations that are being explored (Arnold & Franklin, 2021). If students pose questions that are too general or too deterministic, they may not be able to meaningfully explore the question through data (and educators/designers may not be able to identify the relevant attributes in the dataset, in order to even construct the wrangled data). If they pose questions without considering the potentially different patterns (and thus answers to the questions) that might be revealed during analysis, they may not develop a full understanding of the reciprocal relationship between data and theory. Once they are provided with access to the dataset, we suggest asking students to make predictions and construct hypotheses about what patterns they might find in the data and what those patterns would mean *before* engaging in analysis.

Question-driven selection highlights data analysis as part of a larger cycle of investigation that includes posing questions, collecting and considering appropriate data to address the question, and communicating results. This emphasis on process is echoed in GAISE II (where it is called the "statistical problem-solving process," p. 13, 2020) and in IDSSP ("the basic cycle of learning from data," p. 4, 2019). The emphasis on an initial question, especially if presented as distinct (at least at first) from the available dataset, can support more explicit discussion of hypothesis generation and help guard students against a conflation of data and theory (e.g., Kuhn, 2023). The emphasis on questions can also more deeply engage students in data evaluation and curation, as they examine which attributes are most appropriate to explore.

3.3.2 | Getting it done

Like mathematics-driven selection, question-driven data selection follows the same technical procedures described above for creating groups of attributes that share particular thematic or mathematical characteristics. In this case, the attribute groups would be determined by each attribute's relevance for addressing the question under investigation.

Determining what relevant attributes would be depends on the learning goals of the activity and content area, as well as the questions students are likely to pose. Activities should include sufficient support for students to make predictions ahead of time, so that their investigations move beyond confirmation bias or simple description. If possible, timing activities so that educators have some time after students submit questions to create relevant groups can be highly effective. In our work developing activities to support data analysis in the science classroom, we develop a small set of two to three groups of attributes ahead of time, based on the questions we know students are likely to ask about a given system.

One benefit of question-driven selection versus thematic or mathematics-driven selection is that it can prioritize simple, well-specified questions that can be pursued in depth with a smaller collection of attributes. Consider, for instance, beginning with a simple descriptive question, such as “What industries in California reported toxic release events most frequently in 2023?” using the TRI23CA dataset. A prepared dataset that is targeted to this question might include the “Industry Sector” attribute, with only a few others visible at first. This question can be addressed directly by looking at the frequencies of cases for each industry category (a quick investigation reveals the “Chemicals” and “Petroleum” industry categories report the highest numbers of toxic release events for 2023). However, this initial descriptive investigation can provoke deeper explorations of relationships related to these focal attributes based on the findings. For example, the TRI23CA Python Jupyter notebook demonstrates a follow-up investigation about a secondary question: “Do the ‘Chemicals’ and ‘Petroleum’ industries, as the two most frequent reporters of toxic release events, report the release of specifically *carcinogenic* chemicals at a significantly higher rate than other industries?”

4 | CONCLUSION

While there is growing interest among statistics and data science educators in using large, “messy” datasets, there is less discussion of how these datasets can be wrangled into a pedagogically useful form. Such wrangling can and should be done in principled ways that support the variety of pedagogical goals that a given data exploration activity may be designed to support. Here, we present an organizing framework that can help educators, designers, and even students themselves consider which method of dataset reduction is most appropriate given their pedagogical and investigative goals. The six strategies—three for reducing the number of cases in a dataset, and three for reducing the number of attributes—are summarized in Table 1 below. We also draw some connections between each strategy, statistical problem-solving components in the Guidelines for Assessment and Instruction in Statistics Education (GAISE II; Bargagliotti et al., 2020), and topics in the International Data Science in Schools Project (IDSSP Curriculum Team, 2019) as key contemporary statistics and data science frameworks, which the strategies can especially support.

We hope these resources will help educators and curriculum designers create learning experiences for students that are authentic and engaging, while also maintaining focus on intended learning goals and providing adequate structural and conceptual support to students. We invite readers to use the provided resources to practice their own data wrangling and simplification moves, and consider what opportunities these might open up for students.

CONFLICT OF INTEREST STATEMENT

All four authors have been involved in the creation of the Common Online Data Analysis Platform (CODAP) software or its associated plugins. William Finzer and Tim Erickson have been involved in the development of the tool and Hollylynne S. Lee and Michelle H. Wilkerson have collaborated with them on research grants that support CODAP development.

FUNDING INFORMATION

The work and writing described in this article were supported by the National Science Foundation under award IIS-1900606. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

AUTHOR CONTRIBUTIONS

MHW: Conceptualization, funding acquisition, project administration, resources, software, supervision, visualization, writing – original draft, writing – review & editing. **TE:** Conceptualization, funding acquisition, resources, writing – original draft, writing – review & editing. **HL:** Conceptualization, funding acquisition, writing – review & editing. **WF:** Conceptualization, funding acquisition, writing – review & editing

DATA AVAILABILITY STATEMENT

The data and materials presented as part of this technology demonstration are openly available at http://github.com/CalCoRE/how-to-choosy and within the Open Science Framework at https://doi.org/10.17605/OSF.IO/Z8NS9.

ORCID

Michelle H. Wilkerson https://orcid.org/0000-0001-8250-068X
Tim Erickson https://orcid.org/0000-0001-7066-0758
Hollylynne S. Lee https://orcid.org/0000-0002-7022-2018
William Finzer https://orcid.org/0000-0003-1330-345X

REFERENCES

J. Ainley, R. Gould, and D. Pratt, *Learning to reason from samples: Commentary from the perspectives of task design and the emergence of "big data"*, Educ. Stud. Math. **88** (2015), no. 3, 405–412. https://doi.org/10.1007/s10649-015-9592-4.

P. Arnold, A. Bargagliotti, C. Franklin, and R. Gould, *Bringing complex data into the classroom*, Harvard Data Sci. Rev. **4** (2022), no. 3. https://doi.org/10.1162/99608f92.4ec90534.

P. Arnold and C. Franklin, *What makes a good statistical question?* J. Stat. Data Sci. Educ. **29** (2021), no. 1, 122–130. https://doi.org/10.1080/26939169.2021.1877582.

A. Bargagliotti, C. Franklin, P. Arnold, R. Gould, S. Johnson, L. Perez, and D. Spangler, *Pre-K-12 guidelines for assessment and instruction in statistics education II (GAISE II)*, 2nd ed., American Statistical Association, Alexandria, Virginia, 2020.

D. Ben-Zvi, A. Bakker, and K. Makar, *Learning to reason from samples*, Educ. Stud. Math. **88** (2015), no. 3, 291–303. https://doi.org/10.1007/s10649-015-9593-3.

CalMatters. *Hidden hazards: toxic waste in California* [Article Series]. 2023. https://calmatters.org/series/hazardous-waste-california/.

T. Erickson, M. Wilkerson, W. Finzer, and F. Reichsman, *Data moves*, Technol. Innov. Stat. Educ. **12** (2019), no. 1. https://doi.org/10.5070/T5121038001.

M. Favaretto, E. De Clercq, C. O. Schneble, and B. S. Elger, *What is your definition of big data? Researchers' understanding of the phenomenon of the decade*, PLoS One **15** (2020), no. 2, e0228987. https://doi.org/10.1371/journal.pone.0228987.

R. Gould, *Statistics and the modern student*, Int. Stat. Rev. **78** (2010), no. 2, 297–315. https://doi.org/10.1111/j.1751-5823.2010.00117.x.

IDSSP Curriculum Team. *Curriculum frameworks for introductory data science*. The International Data Science in Schools Project. 2019. https://www.idssp.org/files/IDSSP_Frameworks_1.0.pdf.

S. Kandel, J. Heer, C. Plaisant, J. Kennedy, F. van Ham, N. H. Riche, C. Weaver, B. Lee, D. Brodbeck, and P. Buono, *Research directions in data wrangling: visualizations and transformations for usable and credible data*, Inf. Vis. **10** (2011), no. 4, 271–288. https://doi.org/10.1177/1473871611415994.

S. Kazak, T. Fujita, and M. P. Turmo, *Students' informal statistical inferences through data modeling with a large multivariate data-set*, Math. Think. Learn. **25** (2023), no. 1, 23–43. https://doi.org/10.1080/10986065.2021.1922857.

S. Krishnamurthi, E. Schanzer, D. Schneider, and Z. Drozda. *Datasets for K-12 data science*. n.d. https://ed38e356-3446-418c-9bc9-f0e0d897525a.filesusr.com/ugd/d2c47c_db9901e7a3b04350b561457bea71b48e.pdf.

D. Kuhn. *Defining and developing data literacy*. Routledge Open Research **2** (2023), no. 44, 44.

J. Lane, V. Stodden, S. Bender, and H. Nissenbaum. *Privacy, big data, and the public good: Frameworks for engagement*, Cambridge University Press, 2013. https://doi.org/10.1017/CBO9781107590205.

K. Lanouette, K. L. Cortes, L. Lopez, M. Bakal, and M. H. Wilkerson, *Exploring climate change through students' place connections and public data sets*, Sci. Scope **47** (2024), no. 3, 18–25. https://doi.org/10.1080/08872376.2024.2340444.

S. Miller. *Billboard-Hot-100* [Dataset]. n.d. https://github.com/HipsterVizNinja/random-data/raw/main/Music/hot-100/Hot%20100.csv.

S. Podworny, R. Biehler, and Y. Fleischer, *Young students' engagement with data to create decision trees*, ZDM **57** (2025), 175–191. https://doi.org/10.1007/s11858-024-01649-w.

J. Rosenberg, A. Edwards, and B. Chen, *Getting messy with data: tools and strategies to help students analyze and interpret complex data sources*, Sci. Teach. **87** (2020), no. 5, 30–35.

A. Rubin and J. Mokros, "*Data clubs for middle school youth: engaging young people in data science*," *Looking back, looking forward. Proceedings of the Tenth International Conference on Teaching Statistics, Kyoto, Japan*, M. A. Sorto, A. White, and L. Guyot (eds.), International Statistical Institute, Voorburg, 2018.

E. H. Schultheis and M. K. Kjelvik, *Using messy, authentic data to promote data literacy & reveal the nature of science*, Am. Biol. Teach. **82** (2020), no. 7, 439–446.

D. Weintrop and R. Israel-Fishelson, *Bringing students' lives into data science classrooms*, Harvard Data Sci. Rev. **6** (2024), no. 3. https://doi.org/10.1162/99608f92.6d2aec03.

M. H. Wilkerson and V. Laina, *Middle school students' reasoning about data and context through storytelling with repurposed local data*, ZDM **50** (2018), no. 7, 1223–1235. https://doi.org/10.1007/s11858-018-0974-9.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: M. H. Wilkerson, T. Erickson, H. S. Lee, and W. Finzer, *How to be "Choosy": Wrangling big datasets for the classroom*, Teach. Stat. (2025), 1–21, DOI 10.1111/test.70022